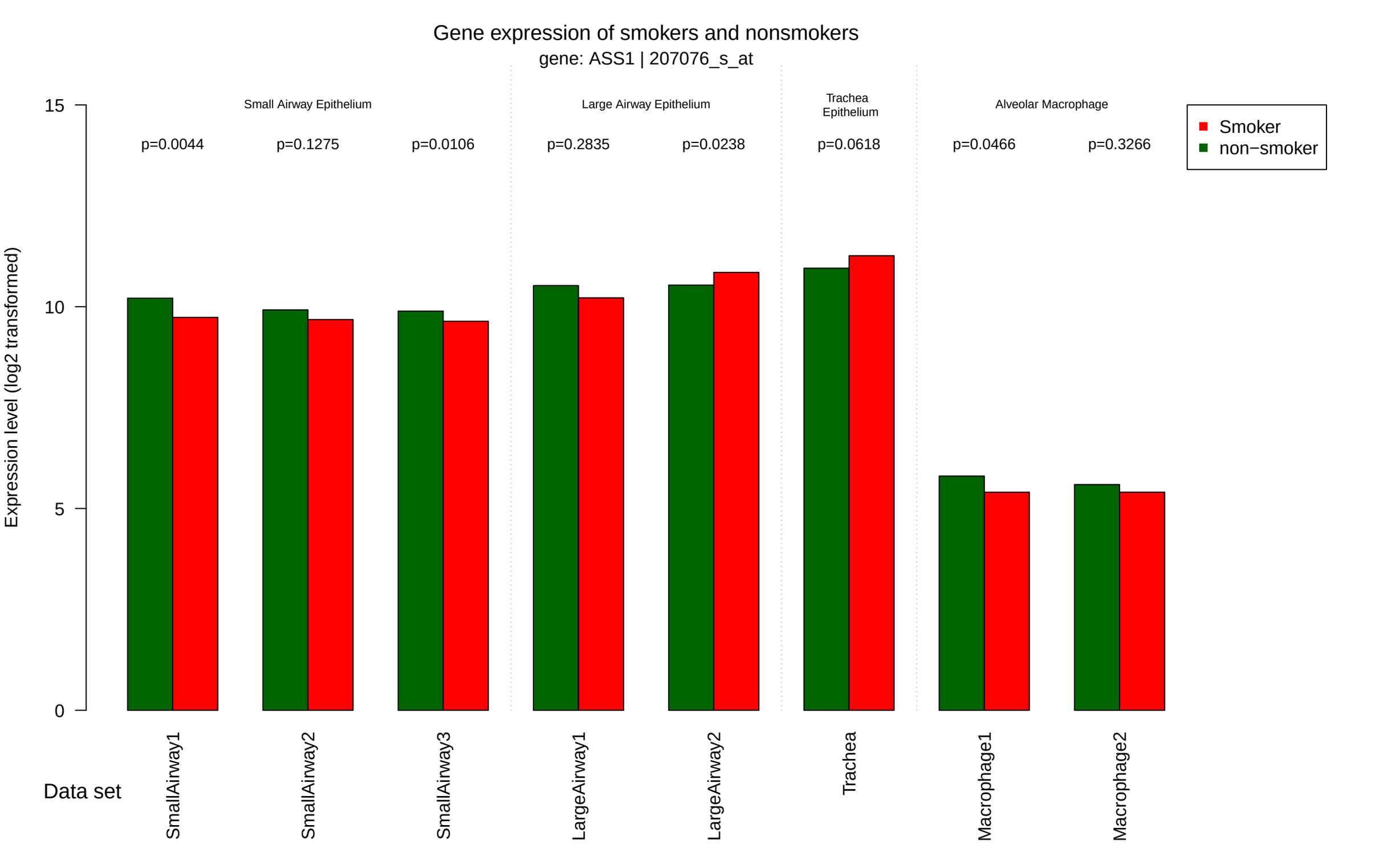
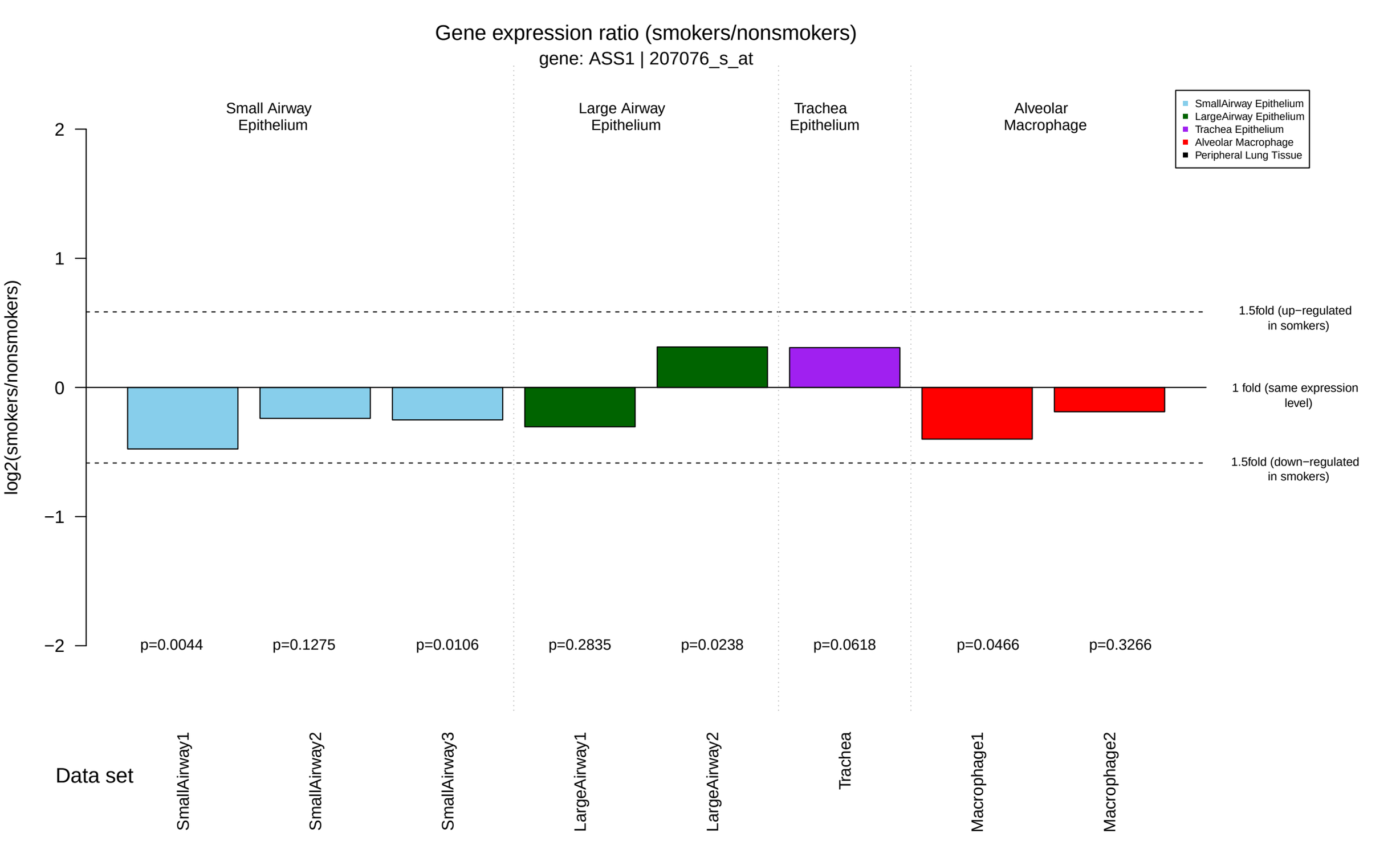
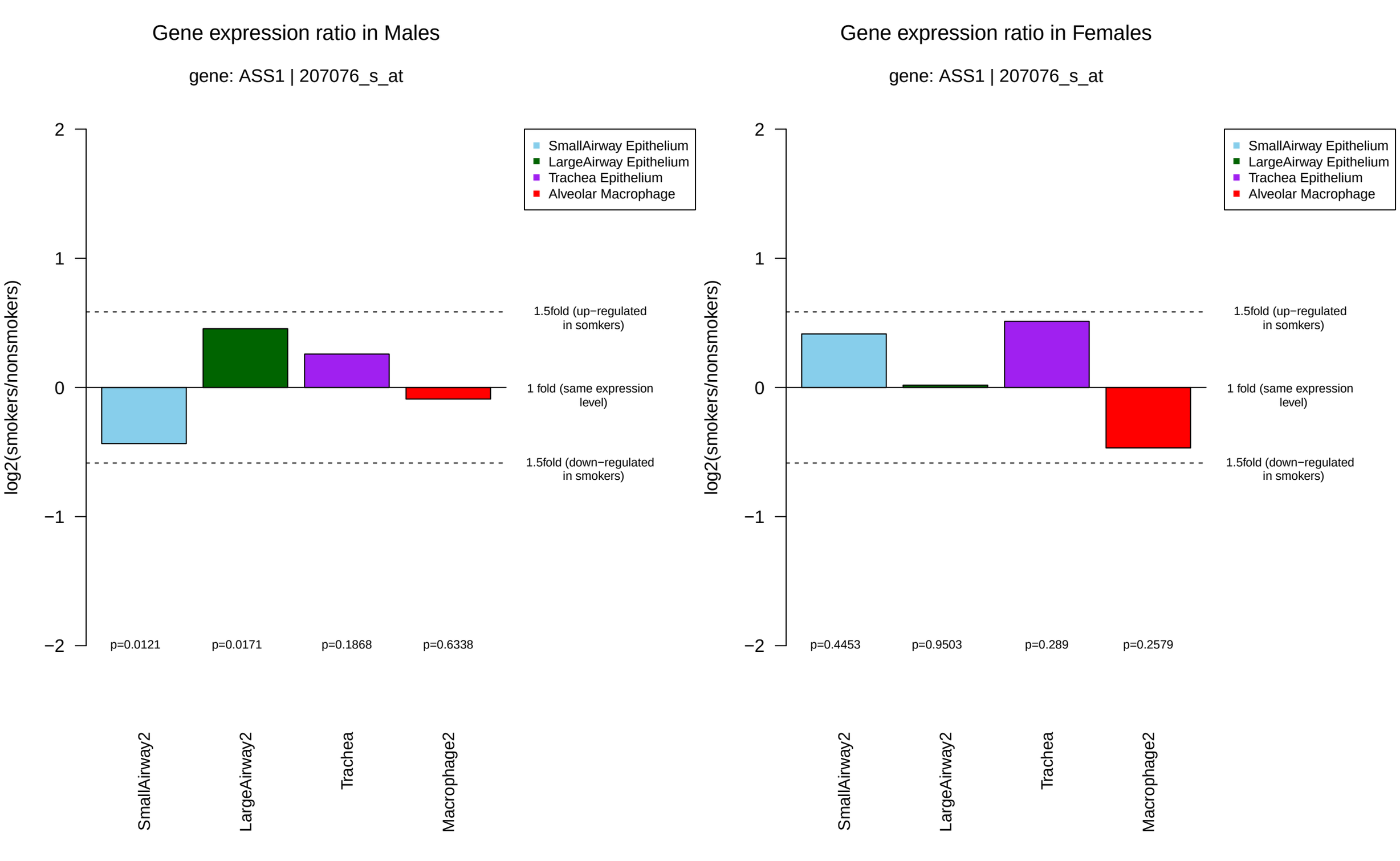
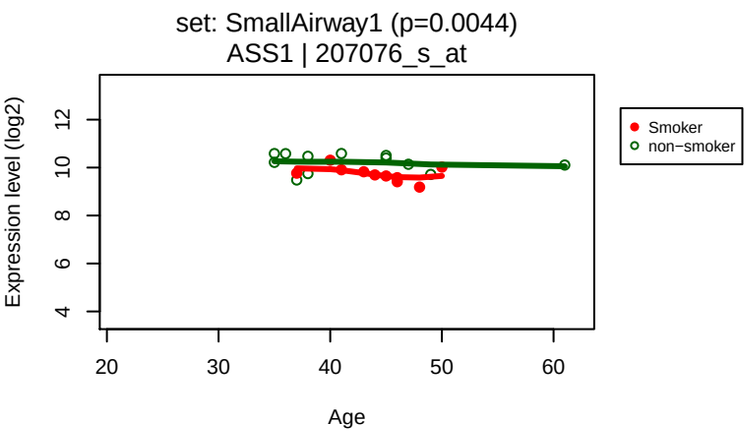
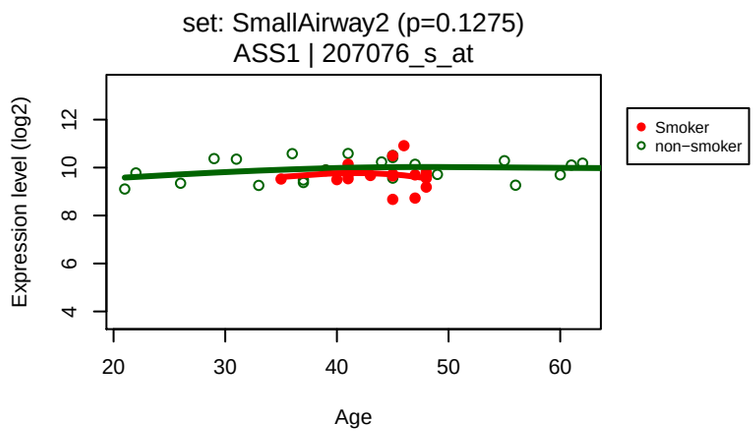
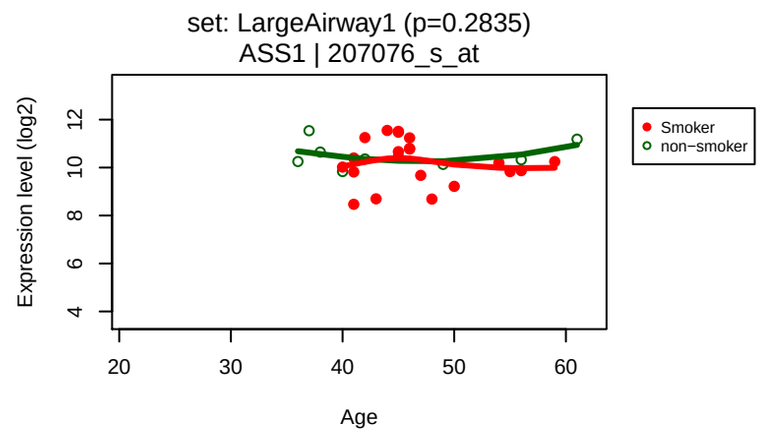

Gene expression of smokers and nonsmokers
gene: ASS1 | 207076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0044
p=0.1275
p=0.0106
p=0.2835
p=0.0238
p=0.0618
p=0.0466
p=0.3266
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

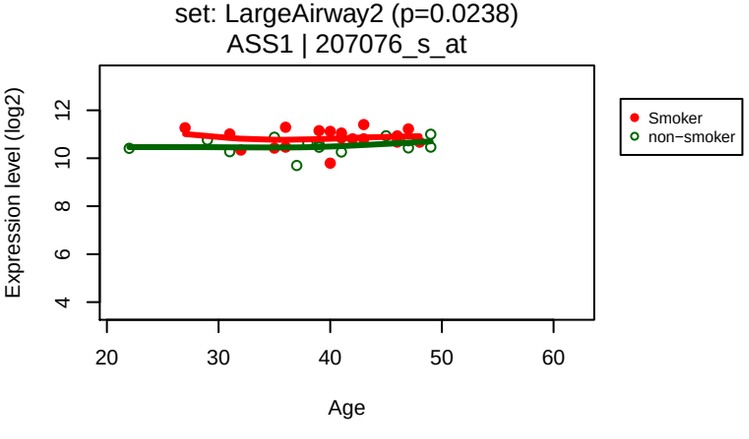

Gene expression ratio (smokers/nonsmokers)
gene: ASS1 | 207076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.1275
p=0.0106
p=0.2835
p=0.0238
p=0.0618
p=0.0466
p=0.3266
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

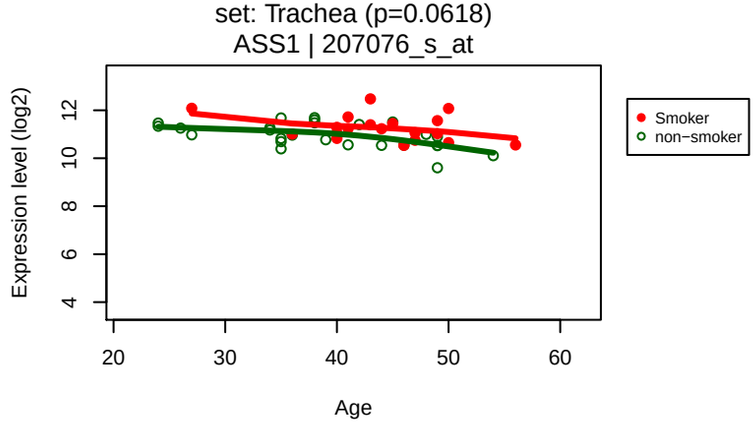

Gene expression ratio in Males
gene: ASS1 | 207076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0121
p=0.0171
p=0.1868
p=0.6338
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASS1 | 207076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4453
p=0.9503
p=0.289
p=0.2579
SmallAirway2
LargeAirway2
Trachea
Macrophage2

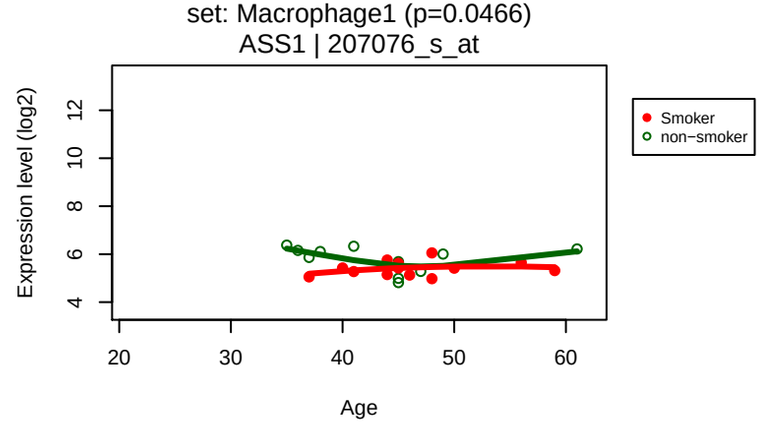

set: SmallAirway1 (p=0.0044)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker

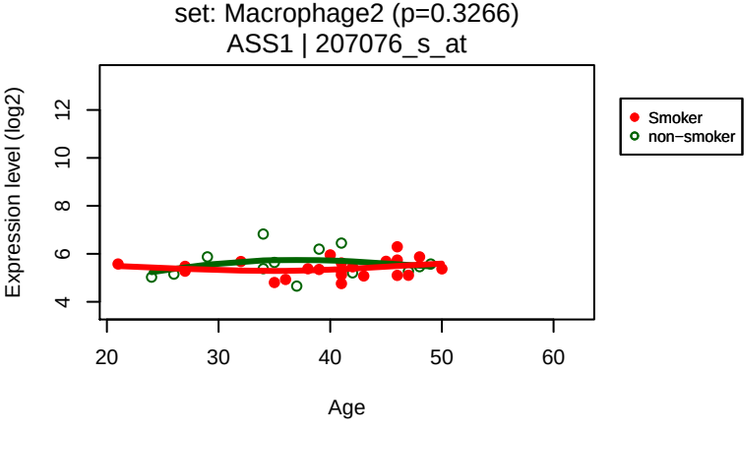

set: SmallAirway2 (p=0.1275)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker

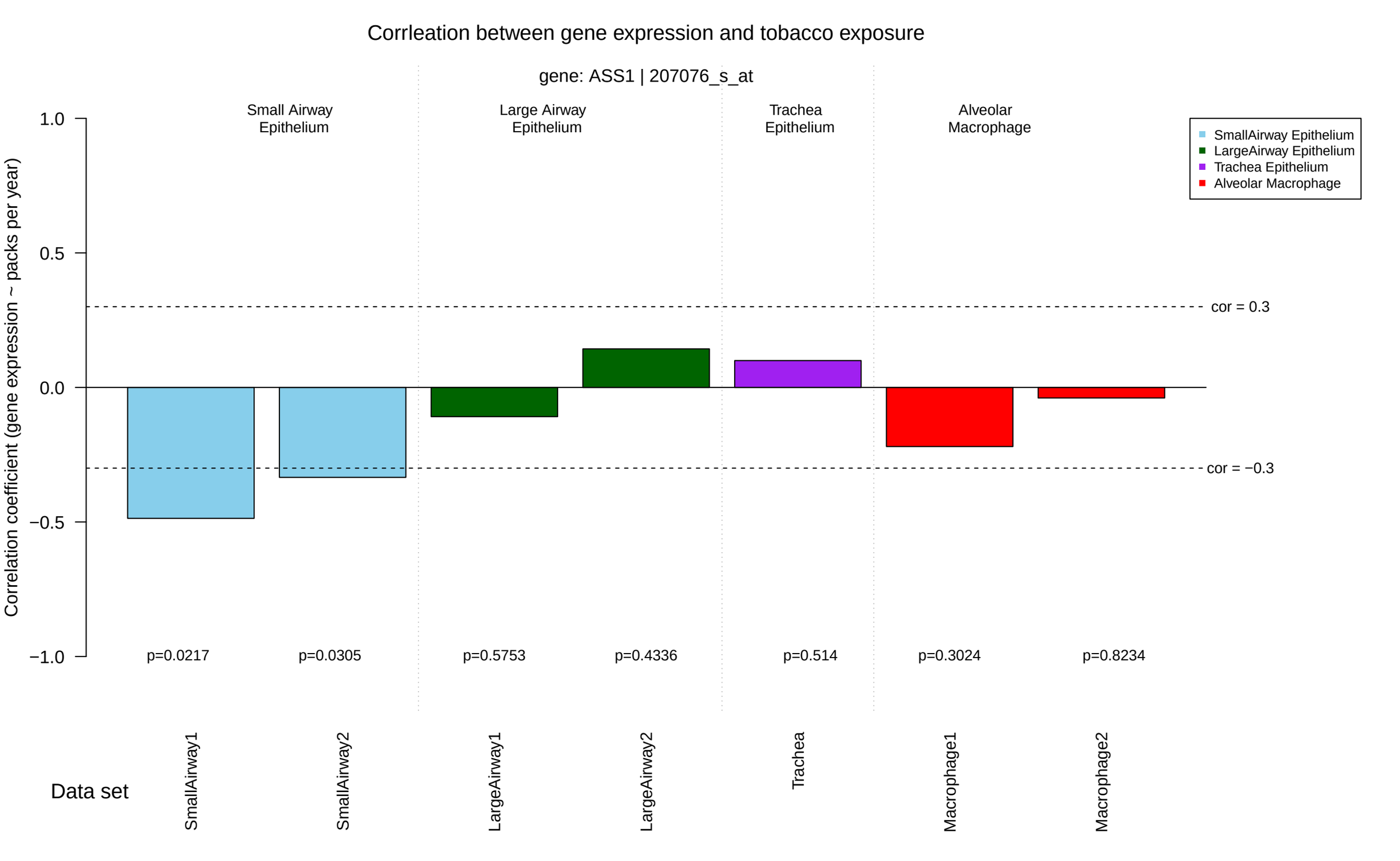

set: LargeAirway1 (p=0.2835)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0238)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0618)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0466)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3266)
ASS1 | 207076_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASS1 | 207076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0217
p=0.0305
p=0.5753
p=0.4336
p=0.514
p=0.3024
p=0.8234
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2